

Predicting Nasdaq Closing Auction Price Movements: A Market Microstructure Study of Order Imbalance, Liquidity and Temporal Dynamics

Introduction

The main question that I wish to answer is as follows:

- What information in the final 10 minutes of the Nasdaq closing auction actually predicts short-horizon relative price movements and how does that information change as the auction approaches the close
- The report is based on the following hypotheses:
- H1 - Auction imbalance carries predictive information about 60-second forward relative returns.
- H2 - Informativeness is not constant, it changes as the close approaches.
- H3 - Cross-sectional and within stock normalization improves generalization relative to raw units.
- H4 - Temporal dynamics (ex: $\Delta\text{imbalance}$) provide greater predictive power than absolute levels.
- H5 - Predictability is dependent upon market regimes such as volatility, liquidity, and imbalance magnitude.

Framework & Results

Understanding the Problem Statement

- [Problem Statement](#)
- [Introduction](#)
- [Nasdaq Closing Cross](#)
- [Orders and the order book](#)
- [Bid-Ask Spread](#)
- [Liquidity](#)
- Mid-price and Weighted Average Price (WAP)
- Order imbalance and Auction imbalance

Dataset Forensics

- Shape: (5237980, 17)
- There are no duplicates in the data since Unique Keys = Number of observations
- Number of Stocks: 200
- Number of Dates: 481
- Columns along with % missingness:

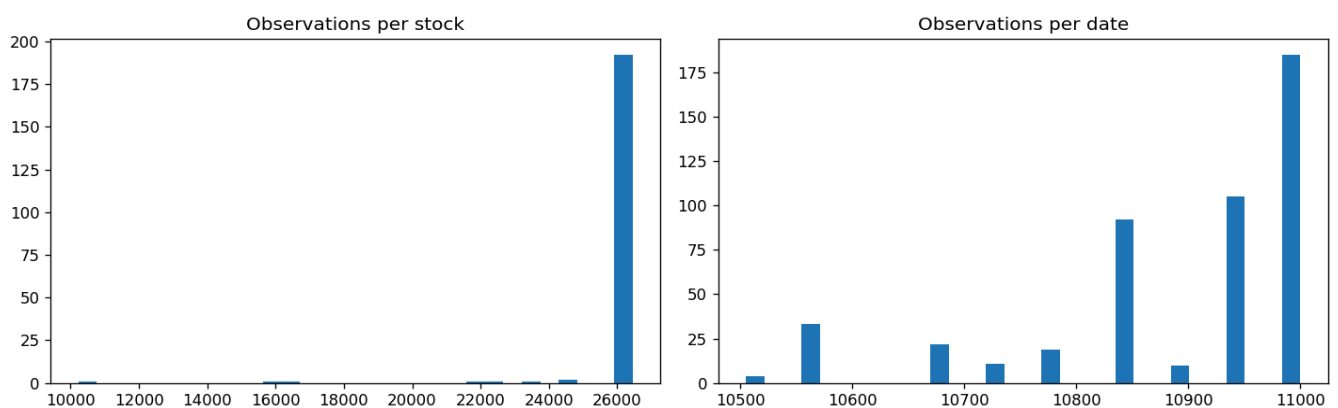
Columns	% missingness
far_price	55.2568356503843
near_price	54.547363678364600
reference_price	0.004200092402032850
imbalance_size	0.004200092402032850
matched_size	0.004200092402032850
ask_price	0.004200092402032850
wap	0.004200092402032850
bid_price	0.004200092402032850
target	0.0016800369608131400
stock_id	0.0
seconds_in_bucket	0.0
imbalance_buy_sell_flag	0.0
date_id	0.0
ask_size	0.0
bid_size	0.0
time_id	0.0
row_id	0.0

- `far_price` / `near_price` are NaN early in the auction since Nasdaq starts to disseminate that after 3:55 pm ET

seconds_in_bucket	near_price missing rate
0	1.0
10	1.0
20	1.0
30	1.0
40	1.0

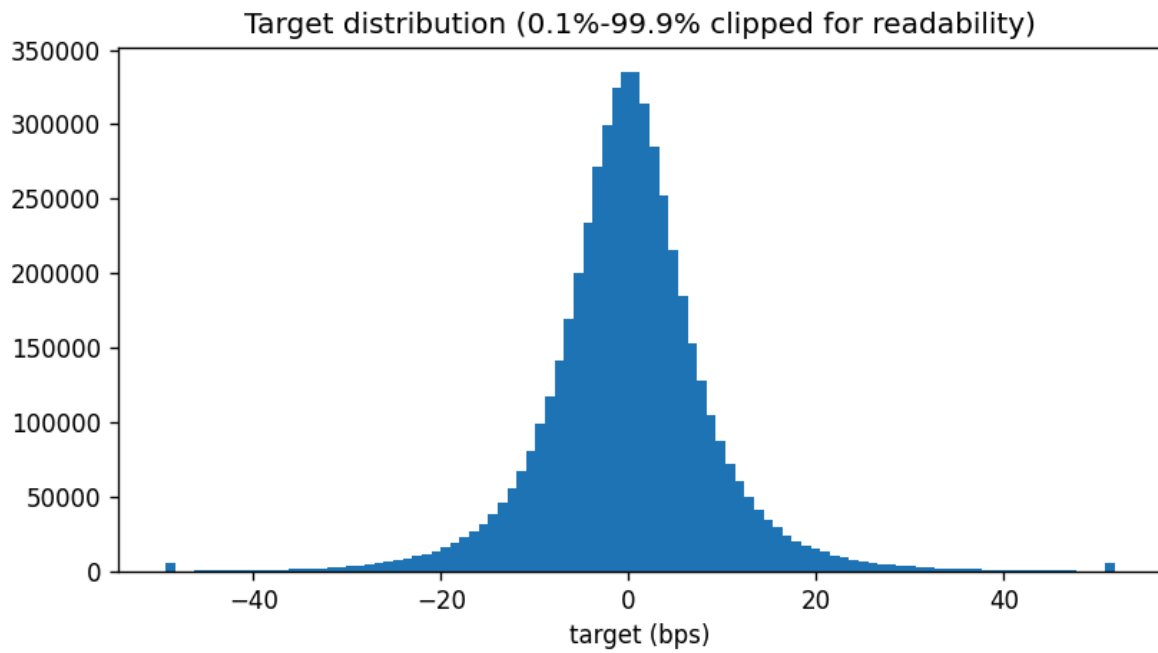
seconds_in_bucket	near_price missing rate
500	0.000042
510	0.000042
520	0.000042
530	0.000042
540	0.000042

- Coverage - number of observations for each stock and each date

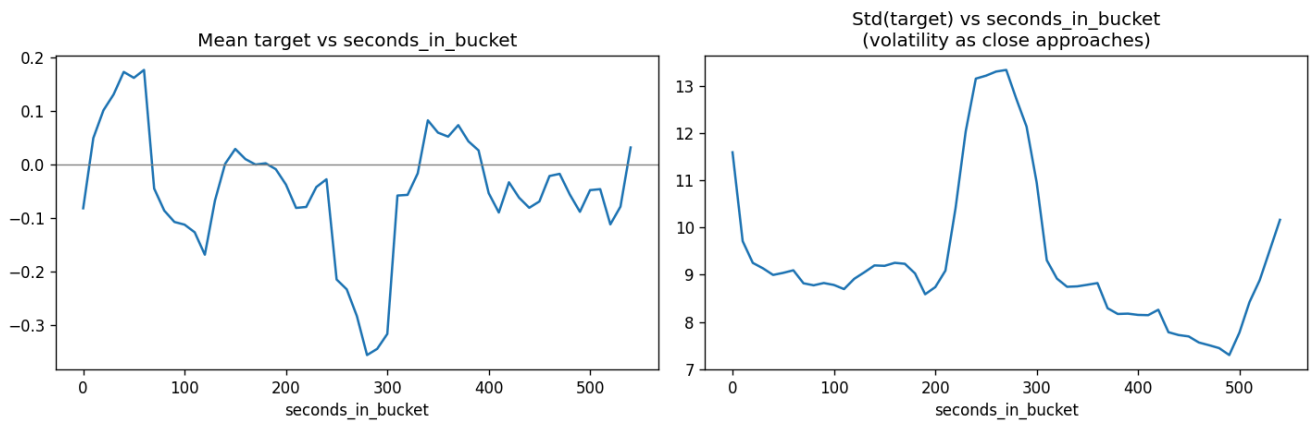


- Distribution of Target

Metric	Value
count	5.237892e+06
mean	-4.756125e-02
std	9.452860e+00
min	-3.852898e+02
25%	-4.559755e+00
50%	-6.020069e-02
75%	4.409552e+00
max	4.460704e+02
skew	0.2047
excess kurtosis	22.5579



- Target vs Seconds in Bucket



- Top 5 stocks with highest mean-target:

stock_id	mean	std	count
156	0.422840	14.086228	24420
67	0.461028	11.051371	26455
180	0.523620	14.413506	26455
92	0.550798	17.907262	26455
11	0.621803	16.080562	26455

- Bottom 5 stocks with lowest mean-target:

stock_id	mean	std	count
153	-0.568705	14.081304	22605
110	-0.524875	6.416165	26455
147	-0.478601	9.525942	26455
25	-0.399342	6.993076	26455
66	-0.395213	7.675980	26455

- Extreme Observations: 56236 (1.074% of rows), bounds = (-31.47, 31.32)
- Feature Distribution

Feature	count	mean	std	min	25%	50%	75%	max
imbalance_size	5237760.0	5.715293e+06	2.051591e+07	0.000000	8.453415e+04	1.113604e+06	4.190951e+06	2.982028e+09
imbalance_buy_sell_flag	5237980.0	-1.189619e-02	8.853374e-01	-1.000000	-1.000000e+00	0.000000e+00	1.000000e+00	1.000000e+00
reference_price	5237760.0	9.999955e-01	2.532497e-03	0.935285	9.987630e-01	9.999670e-01	1.001174e+00	1.077488e+00
matched_size	5237760.0	4.510025e+07	1.398413e+08	4316.610000	5.279575e+06	1.288264e+07	3.270013e+07	7.713682e+09
far_price	2343638.0	1.001713e+00	7.214705e-01	0.000077	9.963320e-01	9.998830e-01	1.003318e+00	4.379531e+02
near_price	2380800.0	9.996601e-01	1.216920e-02	0.786988	9.971000e-01	9.998890e-01	1.002590e+00	1.309732e+00
bid_price	5237760.0	9.997263e-01	2.499345e-03	0.934915	9.985290e-01	9.997280e-01	1.000905e+00	1.077488e+00
bid_size	5237980.0	5.181359e+04	1.114214e+05	0.000000	7.374720e+03	2.196900e+04	5.583168e+04	3.028784e+07
ask_price	5237760.0	1.000264e+00	2.510042e-03	0.939827	9.990290e-01	1.000207e+00	1.001414e+00	1.077836e+00
ask_size	5237980.0	5.357568e+04	1.293554e+05	0.000000	7.823700e+03	2.301792e+04	5.787841e+04	5.440500e+07
wap	5237760.0	9.999920e-01	2.497509e-03	0.938008	9.987810e-01	9.999970e-01	1.001149e+00	1.077675e+00
target	5237892.0	-4.756125e-02	9.452860e+00	-385.289800	-4.559755e+00	-6.020069e-02	4.409552e+00	4.460704e+02
time_id	5237980.0	1.331005e+04	7.619271e+03	0.000000	6.729000e+03	1.334500e+04	1.990700e+04	2.645400e+04

Validation Framework

- Split the data in order of date instead of random splitting so that future market information does not get leaked into training

PAST -----> FUTURE
 TRAIN VALIDATION TEST

- Generate Walk Forward Folds - This is how actual trading strategy would evolve through time

Fold 1:
 TRAIN ----- | VALIDATION

Fold 2:
 TRAIN ----- | VALIDATION

Fold 3:
TRAIN ----- | VALIDATION

Fold 4:
TRAIN ----- | VALIDATION

Baselines

- Evaluation of Prediction
 - Mean Absolute Error (MAE)
 - Root Mean Square Error (RMSE)
 - Correlation - measures whether predictions move with the actual target
 - Directional Accuracy - checks whether predicted return has the same sign as actual return
- Zero Baseline - this predicts 0 for everything
- Historical Mean - this predicts the training period average for every future observation
- Stock Mean - this predicts the average target for each stock
- Time Bucket Mean - predicts the historical average at that point in the auction
- Baseline Results on validation set:

model	MAE	RMSE	corr	directional_accuracy	n
zero	6.487218870501100	9.66440491511357			791945
historical_mean	6.486687955425240	9.664143111501810	1.05960573047625E-17	0.5071488550341250	791945
stock_mean	6.488918615166270	9.665937103408320	0.0074602247074345	0.5052004874075850	791945
time_bucket_mean	6.486476133080710	9.663741014222110	0.010542846091757100	0.506407641944832	791945

- stock_mean is worse than predicting zero, which means that stocks historical average carries no genuine predictive information
- time_bucket_mean edges out the other baselines slightly since it has the best MAE, RMSE and correlation of the four. This is a hint that that average return is not perfectly flat across the auction window and has a small systematic shift

- Model Stability check for walk forward stock mean baseline:

model	MAE	RMSE	corr	directional_accuracy	n
fold_1_stock_mean	6.007615133723950	8.659916984102580	0.00581804581368638	0.4995545911375060	525360
fold_2_stock_mean	6.481176627940740	9.374520217863880	0.007261697910432430	0.5007269329117690	527559
fold_3_stock_mean	6.392043892722180	9.554302276486510	0.0062650788877332200	0.5045397727272730	528000
fold_4_stock_mean	6.188711722351390	9.131177309006590	0.001287094892025710	0.5015124681548270	527945

- There is a bigger swing in MAE and RMSE, which tells that the time period we evaluate on moves the MAE / RMSE more than which baseline you pick.
- This tells us that a single train/val/test number is not trustworthy for comparing real models. Reporting per-fold MAE is more trustworthy for evaluating model performance

Interpretable Microstructure Features

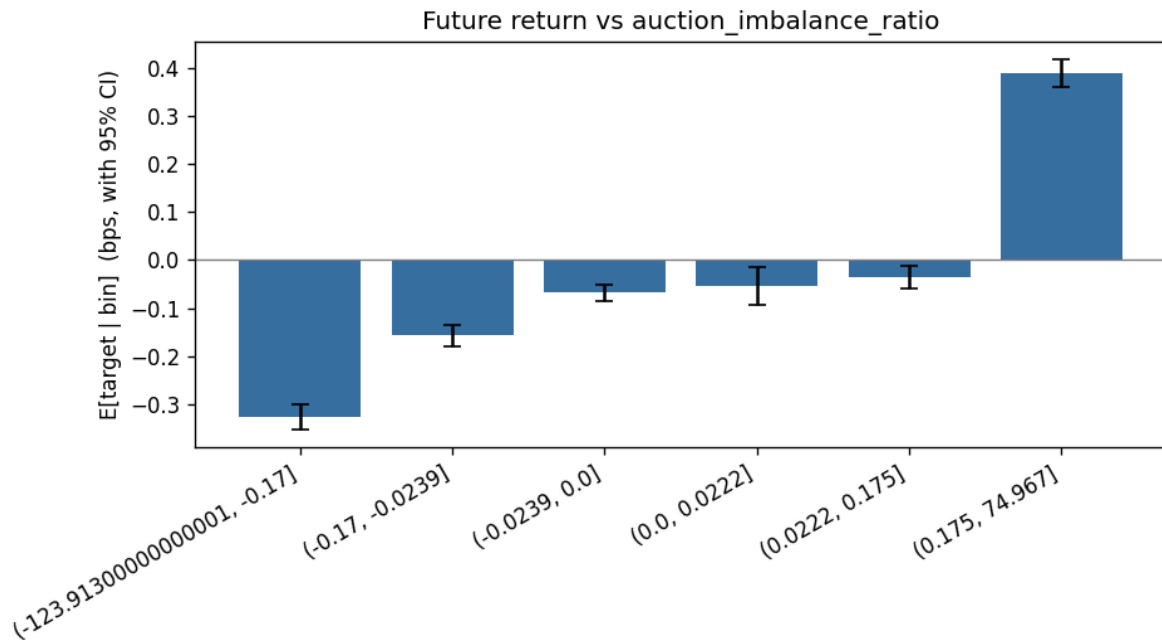
- Mid price = (ask price + bid price) / 2
- Spread = ask price - bid price
- Liquidity imbalance = (bid size - ask size) / (bid size + ask size)
- Market urgency = spread * liquidity imbalance
- Signed imbalance size = imbalance size * imbalance sign
- Auction imbalance ratio = signed imbalance size / (matched size + epsilon)
- Matched size ratio = matched size / (matched size + imbalance size + epsilon)
- Reference Mid distance = (reference price - mid price) / (mid price + epsilon)
- Far Near Distance = (far price - near price) / (mid price + epsilon)

	count	mean	std	min	25%	50%	75%	max
mid_price	3642870.0	0.9999424729 120170	0.0025722306 4366488	0.937371	0.9986740000 000000	0.999922	1.001135	1.052754
spread	3642870.0	0.0005407923 033212820	0.0005234003 169020230	2.999999999 97524E-06	0.0002230000 0000008400	0.0003830000 000000220	0.0006720000 000000060	0.01617100 0000000000
liquidity_imbalance	3642980.0	-0.014069858 036847200	0.5848735788 918470	-0.99991265 36968480	-0.500380901 0383170	-0.003652733 8482985000	0.4700346754 1936100	0.99994014 19283010
market_urgency	3642870.0	-7.056433725 34049E-06	0.0004503002 9759711100	-0.01210104 6985136000	-0.000164956 8375751930	-1.530155260 42782E-06	0.0001540451 1928718300	0.01353606 3706686000
signed_imbalance_size	3642870.0	-137760.8122 514390	20020813.742 595400	-1192693599 .62	-1104839.887 5000000	0.0	978008.0225	1228661266 .7
auction_imbalance_ratio	3642870.0	0.0061859903 28202390	0.6795085616 402430	-123.912004 72226300	-0.076846335 52796980	0.0	0.0764135899 9347730	74.9666697 9036470
matched_size_ratio	3642870.0	0.8638375283 702280	0.1682873317 2102300	0.008005635 665070470	0.7954036794 361020	0.9288230420 038780	0.9969277987 683530	1.0
ref_mid_dist	3642870.0	-2.041267567 66687E-07	0.0003115115 2955353700	-0.01004408 9525777000	-0.000140651 12949798200	0.0	0.0001402323 2926857100	0.02002703 8084817200
far_near_dist	1628479.0	0.0029097748 03573070	0.8590591366 320040	-0.98711982 8237499	-0.000303633 21080236400	0.0	0.0002118170 428180870	438.336336 91786200

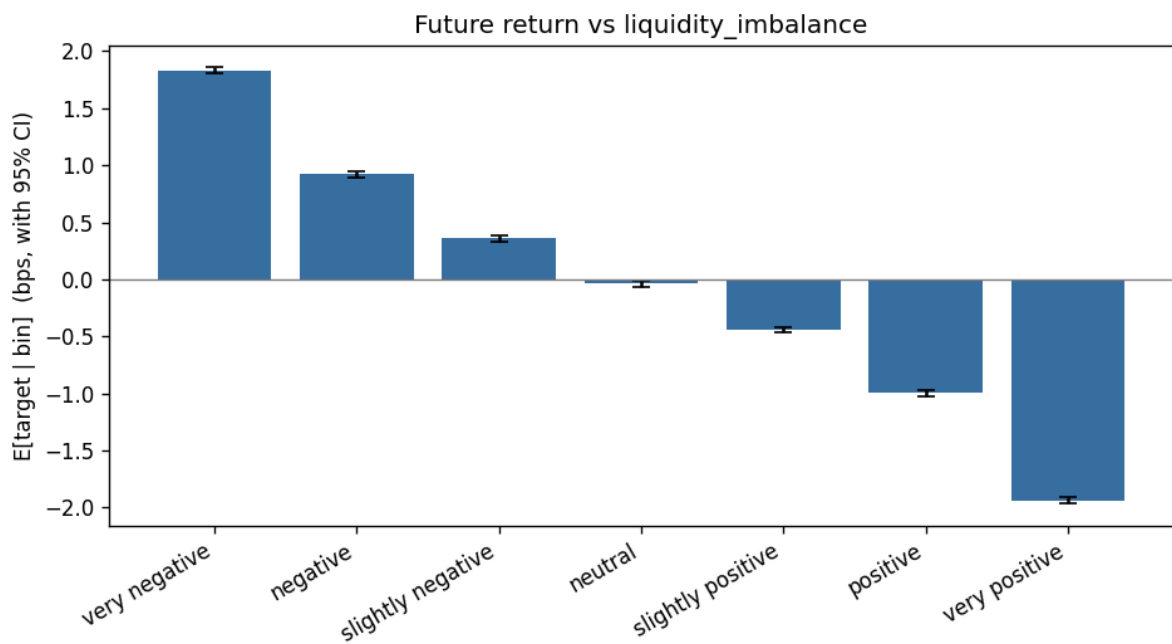
H1 - Does Order Imbalance contain statistically significant information about the target?

- Equal Frequency (Quantile) binning creates bins such that each bin contains roughly the same number of data points
- We analyze signal vs target for auction imbalance, liquidity imbalance, market urgency and match sized ratio
- Grouping by the bins - below are the values calculated for each bin
 - Mean
 - Standard deviation (std)
 - Count (n)
 - Standard error of the mean (sem) - measures how far sample mean is likely to be from the true population mean = $\text{std} / (n)^{0.5}$
 - 95% confidence interval - a range of values that is 95% likely to contain the true population parameter = $[\text{mean} - 1.96 * \text{sem}, \text{mean} + 1.96 * \text{sem}]$
 - Feature min and feature max

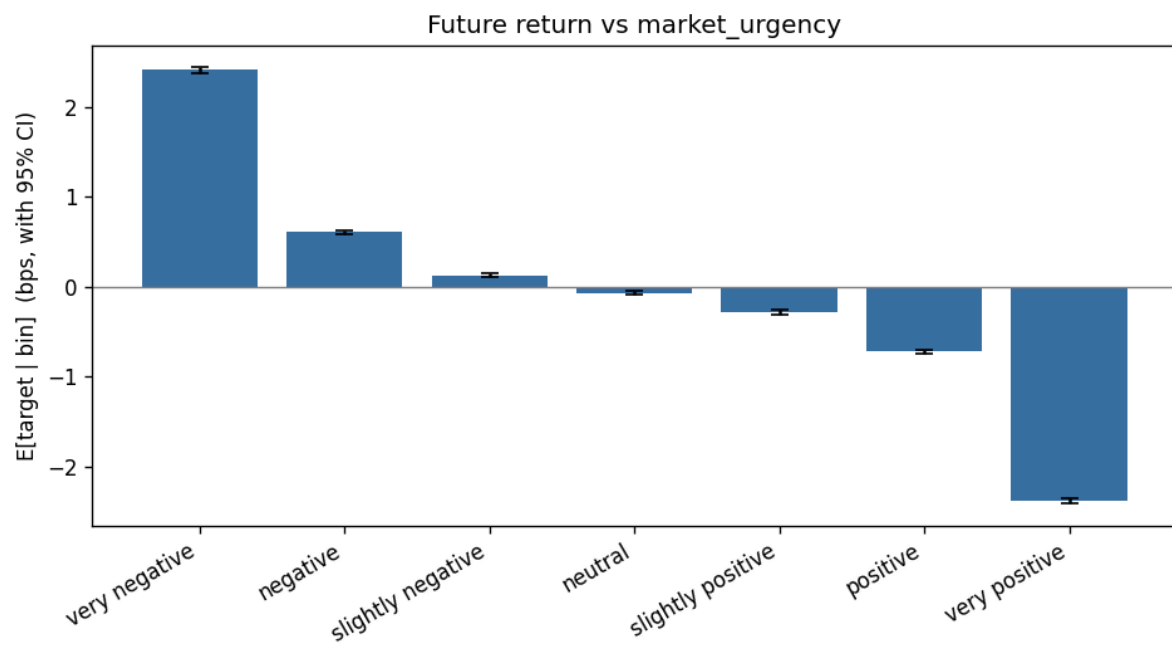
- Spearman correlation - measure the strength and direction of a monotonic relationship
- Pearson correlation - measures the strength and direction of a linear relationship
- ANOVA - a method used to test if mean values of three or more independent groups are significantly different from each other
- Plots of $E[\text{target} \mid \text{signal bin}]$ along with error bars
 - Auction Imbalance Ratio



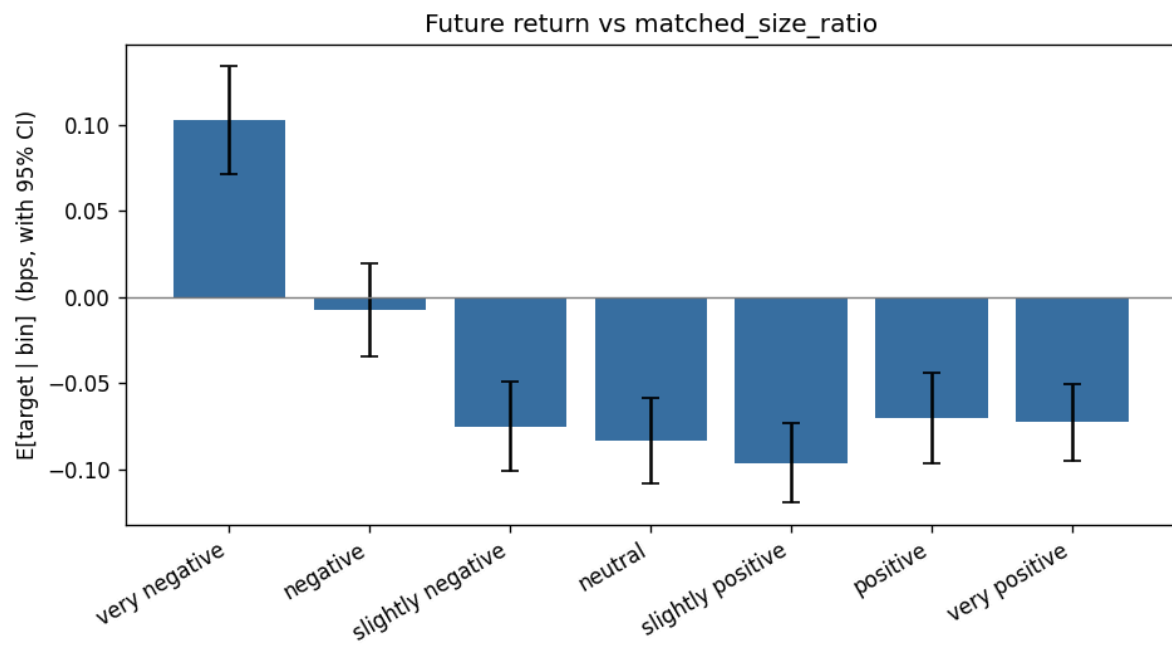
- Liquidity Imbalance



- Market Urgency



- Matched size ratio



- Significance Summary

feature	n	spearman_r	spearman_p	pearson_r	pearson_p	anova_f	anova_p
liquidity_imbalance	3642948	-0.13530508395932800	0.0	-0.121120573924244	0.0	8921.582939861970	0.0
auction_imbalance_ratio	3642870	0.021942477285091600	0.0	0.011925910140091500	1.06797663770864E-114	372.1638558357410	0.0
matched_size_ratio	3642870	-0.0022290109723308700	2.09645974168422E-05	-0.007754536019809060	1.44835500052324E-49	28.131560060152500	8.13873386968299E-34
market_urgency	3642870	-0.14743427609349400	0.0	-0.1639215870946460	0.0	12053.329896661800	0.0

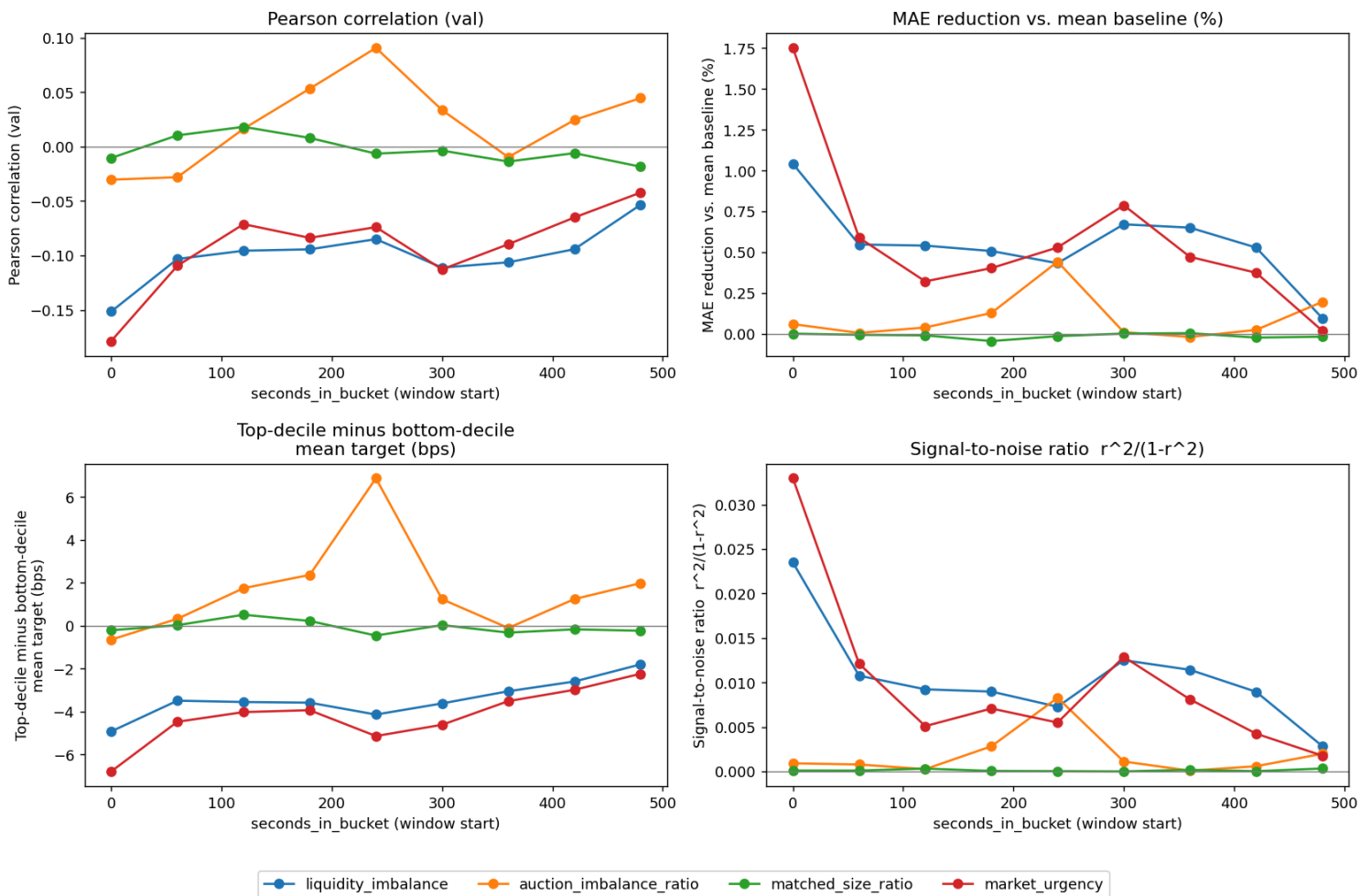
- Yes, order imbalance does predict the 60 second future return but not all types of imbalance give the same information
- The more buying pressure was sitting in the book, the more the price actually tends to fall over the next 60 seconds, and the more the selling pressure is there, the more it tends to rise. Moving from the most negative to the most positive imbalance group swings the average future return by about 3.8 basis points, and the pattern moves in a clean, consistent step from one group to the next.
- Market Urgency (spread * liquidity imbalance) shows an even bigger version of the same pattern with a swing of about 4.8 basis points.
- The likely explanation is in how the dataset's WAP is built, it weights the bid price by the ask size, and the ask price by the bid size. That means a book already tilted toward more bid size has pulled the current WAP up before we even look at the future return. What we're seeing 60 seconds later is largely that price correcting back down
- More unmatched buy side interest in the auction is followed by higher future returns. Unmatched auction demand is real, unfilled buying or selling pressure that still has to be absorbed before the close, so it keeps pushing the price the same direction rather than reverting. The effect is real but much smaller, swinging the average future return under 1 basis point only.
- The ratio of matched to unmatched auction size, showed no meaningful relationship at all. The pattern across its groups is flat with no consistent direction. As the sample size is very high, the fact that the feature is statistically significant alone does not tell much. What matters is the size and consistency of the effect.

H2 - Does Predictability increase as the close approaches?

- Divide the seconds_in_bucket into fixed width windows with size of 60 seconds
- For each time window and each signal (auction imbalance, liquidity imbalance, market urgency, match size ratio) we calculate the following metrics on the validation window
 - Spearman correlation
 - Pearson correlation
 - Model MAE: a OLS is fitted on the train window and scored on the validation window
 - Baseline MAE: a constant train mean baseline scored on the same validation window as Model MAE
 - MAE reduction: $\% \text{ reduction of Model MAE relative to Baseline MAE} = 100 * (\text{baseline MAE} - \text{model MAE}) / \text{baseline MAE}$

- Decile spread: difference in the target for the strongest positive signal observations and strongest negative signal observations = mean of target | top decile of signal - mean of target | bottom decile of signal
- Signal to Noise ratio (snr): it is the ratio of the variance explained by the signal to the residual (unexplained variance) = $r^2 / (1 - r^2)$, where r is the validation set pearson r
- Plot of correlation, MAE reduction %, decile spread, signal to noise ratio for each signal vs window start

RQ2 - Information content of imbalance signals vs. time-to-close



signal	early_abs_r_mean	late_abs_r_mean	early_snr_mean	late_snr_mean
liquidity_imbalance	0.11118428599154300	0.09002734590836480	0.013119380207776100	0.008608789759467600
auction_imbalance_ratio	0.03200856054942170	0.040573628620759200	0.0012048976027776400	0.0024182448557737100
matched_size_ratio	0.011707542225850000	0.00967799995641356	0.00015186555732580300	0.00012442958619026000
market_urgency	0.11077781024405100	0.07675536644508240	0.014312209501321100	0.0065079913016464600

- Conclusion
 - The answer depends on which imbalance measure we see and neither one moves in a simple straight line
 - The book based measures (liquidity imbalance and market urgency) are actually at their most informative right at the start of the ten-minute window. For these two signals, predictability decreases toward the close
 - The auction based measure (auction imbalance) starts out weak in the first two minutes, then converts into a positive signal by around the two-minute mark, and builds to its strongest point of the whole session between roughly minute four and minute five. After that peak it fades again through minutes five to seven, then picks back up somewhat in the final two minutes. Comparing the first half of the window to the second half on average, this signal does get modestly stronger later on — but it's a bump in the middle-to-late window, not a smooth climb all the way into the close.
 - The timing closely lines up with Nasdaq's official imbalance updates switch from every 5 seconds to every one second
 - The matched-size ratio, which showed no real signal in H1, stays flat and uninformative at every stage of the window — there's no point in time where it becomes useful.

H3 - Model 1: Ridge Regression

- Feature Set and Feature Preparation
 - Raw Features - original features from the optiver dataset
 - Imbalance Features - derived from raw features earlier
 - Time Features - this contains `second_in_bucket` and `is_late_auction` which flags the post 300s regime (motivated from the finding of H2)
 - Cross Sectional Features - features that express each stock's signal relative to other stocks at the same date and time: we calculate a cross-sectional z-score and percentile rank for liquidity imbalance, market urgency and auction imbalance ratio
 - Relative z-scores - it normalizes each stock signal against its own history, which indicates whether the signal is unusual relative to the stock's own history
 - We drop the rows in which the target is missing
 - We fill the missing values of training data with the median of the training data and missing values of validation data with the same median of the training data.
 - Optional standardization: $z = (x - \mu) / \sigma$, the train and validation set are standardized with the same mean and std calculated from the training data
- Training and Evaluation
 - Fit a ridge model on the training data to learn the coefficients of features
 - Predict the value of target for the validation data
 - Evaluate the prediction by calculating the MAE, RMSE, correlation and directional accuracy
- Experiments - we run the model on the following feature sets
 - Raw Features without standardization
 - Raw features with standardization
 - Imbalance Features with standardization
 - Time Features with standardization
 - Cross Sectional features with standardization
 - Features relative to stock's history with standardization
 - Features relative to stock's history at that time in the auction (`second_in_bucket`)

- Results

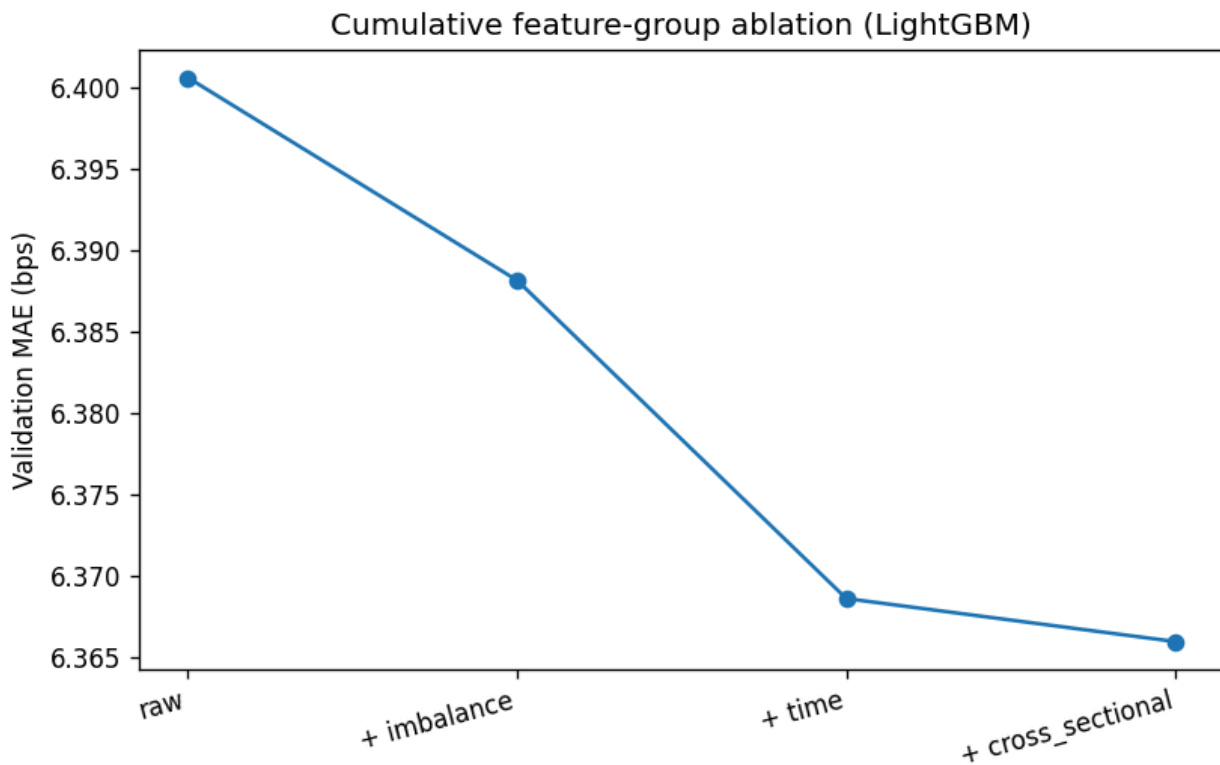
model	MAE	RMSE	corr	directional_accuracy	n
raw	6.46705952543871	9.63857350575269	0.07917494800772150	0.5290771455088420	791945
normalized	6.44112729899227	9.633398224420820	0.10802487916327800	0.5411007077511700	791945
imbalance	6.441945122910410	9.633948970362140	0.10697218916039500	0.5407761902657380	791945
time	6.486026235784770	9.663572195213600	0.011447375117259400	0.5084014672736110	791945
cross_sectional	6.443112771309380	9.627970830637130	0.10163414666916900	0.5412749622764210	791945
stock_relative	6.487901367007650	9.665828070539870	-0.0015083405676104300	0.5024414574244420	791945
bucket_relative	6.487385851720480	9.664200965770520	0.005633458651579200	0.5045274608716520	791945

- Normalizing the raw features improves the model significantly, which is a well established fact for ridge regression. If the features are non-standardized then ridge regression penalizes the features unevenly.
- The 6 imbalance features give almost the same result as normalized raw columns give which shows that nearly all the linear predictive value sitting in the raw order-book and auction columns is captured by the six compact, interpretable ratios
- The cross sectional features also performed well producing the lowest RMSE and highest directional accuracy.
- time , stock relative and bucket relative performed poorly when used alone. The correlation is very poor, suggesting that these features do not provide much predictive information by themselves to the ridge model.
- Coefficients of features - find them in this link [Model 1](#)
 - Normalized Raw Features - Since wap, bid_price, ask_price and reference_price are all related (multi-collinearity) to each other, their individual coefficients cannot be used to draw any conclusions. Their coefficients swing to large, sign alternating values that mostly cancel out.
 - Imbalance Features - Market urgency dominates with a strong margin. Since market urgency is derived from liquidity imbalance, it suppresses the coefficient of liquidity imbalance. This combined with H1 findings shows that market urgency is the more complete version and liquidity imbalance does not add much if market urgency is already there. Reference mid distance (distance between reference price and mid price) also shows up as an important feature which was not explored earlier.
 - Cross Sectional Features - Here also market urgency has the strongest coefficient followed by liquidity imbalance. The cross sectional liquidity imbalance rank has a positive coefficient which indicates that relative liquidity information contributes meaningful signal

H3 - Model 2: LightGBM

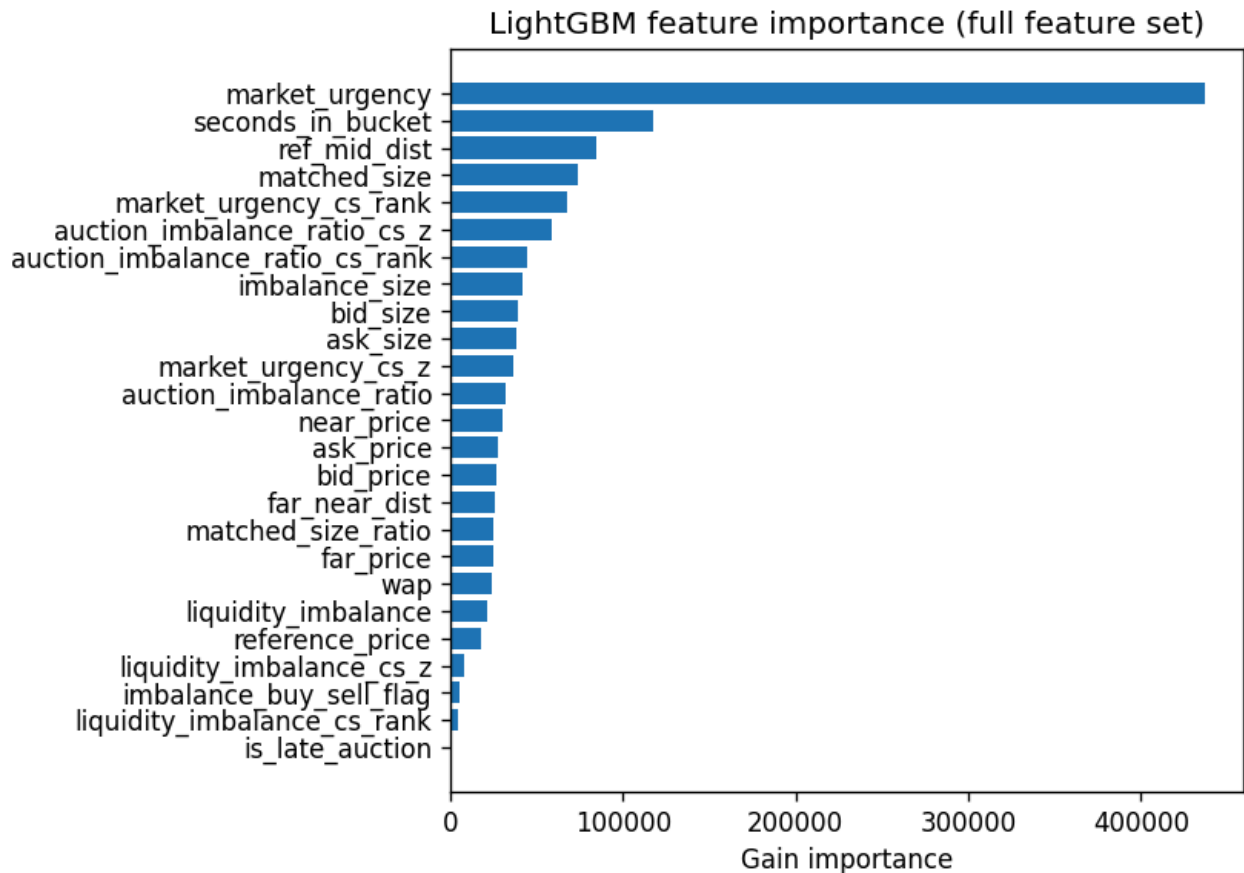
- We keep on adding one feature group at a time from the above created feature groups and retrain the model. We train and evaluate all the models and calculate change in MAE which gives each feature groups sequential marginal contribution
- [LightGBM](#) handles NaN and hence doesn't require median-imputation
- Configuration
 - objective: loss function to be optimized during training = regression_l1
 - learning rate: step size at each iteration during gradient descent = 0.05

- `n_estimator`: this specifies the number of trees to be built = 500
- `n_leaves`: specifies the maximum number of leaves in each tree = 63
- `early_stopping_rounds`: if validation performance does not improve for the specified number of rounds training stops = 50
- Training and Evaluation
- Results



- Adding features gradually improves the prediction performance. Starting from the raw features, the MAE decreased by 0.54 % upon adding all the features.
- The time features provide the largest marginal improvement which indicates that time features carry predictive information when combined with other features and do not contain much predictive information alone.

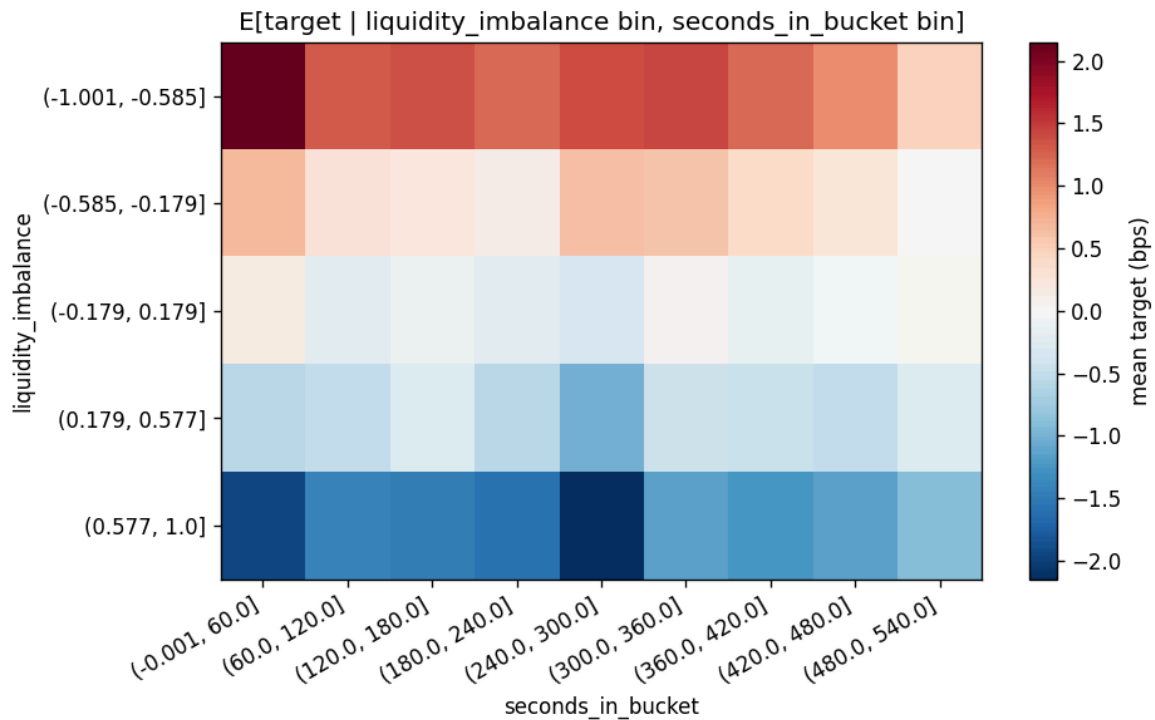
- Feature Importance - gives information about which features produced the greatest reduction in loss: LightGBM gain tells you which features were useful for tree splits



- The gain-based feature importance of the full model reinforces the findings from model 1, showing that market urgency is the most important feature by a large margin and reference mid distance is also a significant feature ranked third in feature importance.
 - Seconds in bucket emerges as the second most important feature showing that even though seconds in bucket did not contain predictive information when used alone, when it is used along with other features, it provides significant predictive information.
 - Looking at the feature groups as a whole, the imbalance related features have the highest gain importance followed by raw features, cross-sectional features and time features
- Interaction Experiment - Does the predictive effect of imbalance depend on time
 - Model A: no interaction between signal and time
 - Model B: Explicit interaction (Signal * Time)
 - Model C: LightGBM ~ signal + time, it will discover non linear relations automatically

model	MAE	RMSE	corr	directional_accuracy
A_additive_only	6.4545540284145400	9.62002748455867	0.09750749119047770	0.5396081798609750
B_explicit_interaction	6.452403563966380	9.618221258248820	0.09958967870906200	0.5397003579794050
C_lightgbm_2feature	6.441704782050500	9.611849558040810	0.10436712246836800	0.5409037243747990

- The results improve slightly upon adding an explicit interaction term to the additive only model. The LightGBM model performs better and further improves the metrics
- The results suggest that interaction is useful, but the relationship is more complex than a single product term can capture



- The pattern across the liquidity-imbalance bins is very strong and consistent across time. The lowest liquidity-imbalance bin has positive average target values in every time interval, whereas the highest liquidity-imbalance bin has negative average target values throughout. The middle bin is close to zero in most periods.
- The magnitude of this relationship changes with the auction stage. The time effect itself is not monotonic which explains why introducing an explicit interaction term (Model B) did not explain the target like LightGBM did.
- The signal is particularly strong in the first minute when Nasdaq first begins early dissemination of closing information. A possible reason behind this is that the market immediately reacts to the initial revelation of the auction book's state.
- The signal decays thereafter and again becomes strong at the 300 second mark. Nasdaq stops taking MOC orders after 300 seconds and starts disseminating the NOII every second along with near indicative clearing price and far indicative clearing price, the strong signal might be because of that

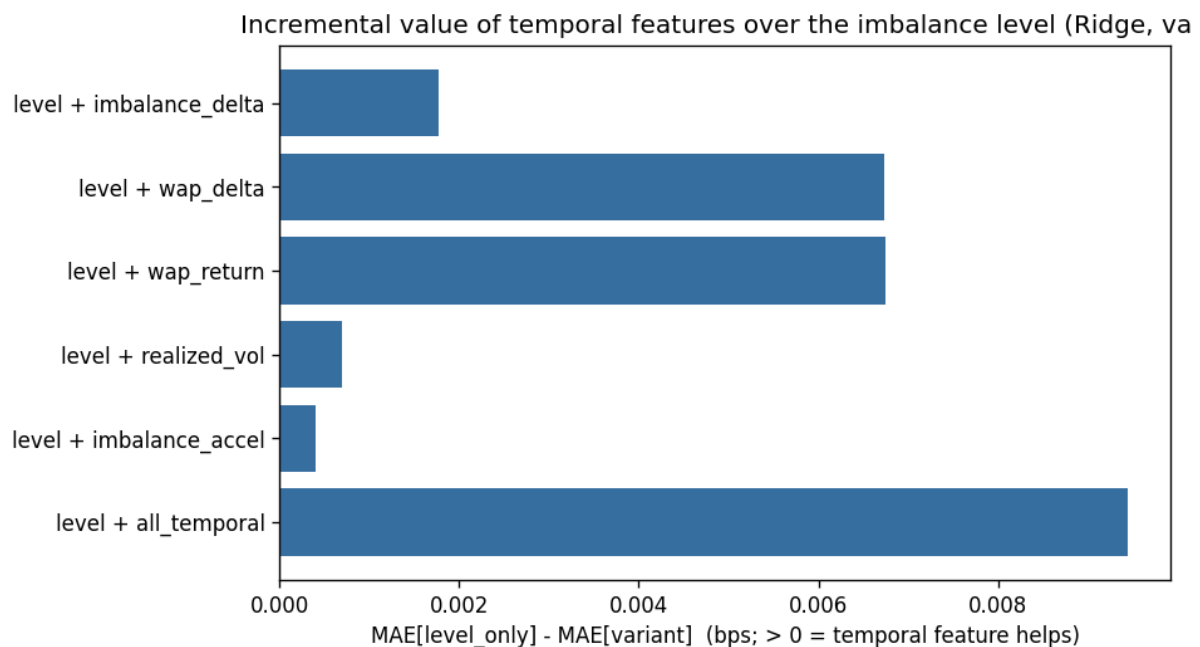
H4 - Temporal Dynamics: Is the absolute level of imbalance predictive, or is the change in imbalance itself informative?

- We calculate the following:
 - Change in imbalance = imbalance at time t - imbalance at time $(t - k)$, where we have taken k to be 10 seconds
 - Change in wap = wap at time t - wap at time $(t - k)$, where we have taken k to be 10 seconds
 - Return in wap = Change in wap / Original wap
 - Rolling Volatility of wap = std of percentage change in wap for the past 6 rows
 - Second Order change in imbalance = change in imbalance at time t - change in imbalance at time $(t - k)$, where we have taken k to be 10 seconds
- Raw Signal Strength - we use auction imbalance ratio and the above features and analyze their relationship with target using the same function used in RQ1

feature	spearman_r	spearman_p	pearson_r	pearson_p	spearman_r_val	top_minus_bottom_bin_bps
auction_imbalance_ratio	0.021942477285091600	0.0	0.011925910140091500	1.06797663770864E-114	0.029745890874626000	0.7144093014381920
imbalance_delta	0.030592078188479300	0.0	0.020574021268025500	0.0	0.03232246842562460	0.8975502074129680
wap_delta	-0.06260074364161860	0.0	-0.07273806491503860	0.0	-0.038464290621900300	-1.991066827076500
wap_return	-0.06260226131961140	0.0	-0.07271565856264790	0.0	-0.03846474633798980	-1.9912691348463700
realized_vol	0.007220994196697080	1.0476985522306E-41	0.01193551573440280	8.97333703084749E-111	0.010914201808894100	0.31347867152989500
imbalance_accel	0.01639262902323230	3.54610740928986E-207	0.010309552525284200	3.89829159835017E-83	0.015844555405777000	0.47818962832712700

- The change in imbalance has a stronger relationship with the target than the level of imbalance (spearman correlation 0.031 vs 0.022). This is also consistent on the validation data (0.032 vs 0.030)
- The recent move in wap has the strongest relationship with the target and it is negative. If wap goes up in the last 10 seconds, the target tends to come down over the next 60 seconds and the other way round. This is because of the same wap effect explained during H1.
- Realized volatility and the second order change in imbalance are very weak on their own.
- We fit a ridge for the following cases and evaluate on validation data
 - Level Only
 - Level + imbalance delta
 - Level + wap delta
 - Level + wap return
 - Level + realized volatility
 - Level + imbalance acceleration
 - Level + all temporal features

model	MAE	RMSE	corr	directional_accuracy
level_only	6.485896046056630	9.663184041564290	0.014043820215997400	0.5109104798944370
level + imbalance_delta	6.484128355999250	9.659964256466690	0.02962275227994660	0.5141846971696270
level + wap_delta	6.479161894106920	9.660087028841110	0.0391834801158012	0.5156342927854840
level + wap_return	6.479154655427460	9.660043935838670	0.03923910091013110	0.5156431317831420
level + realized_vol	6.4852010813371500	9.661402738752990	0.024732126051979900	0.5105341911370110
level + imbalance_accel	6.485494394617850	9.66299442585885	0.015453405573223100	0.5110910479894440
level + all_temporal	6.476448231168570	9.65454518406808	0.04941577742888810	0.5192469173995670

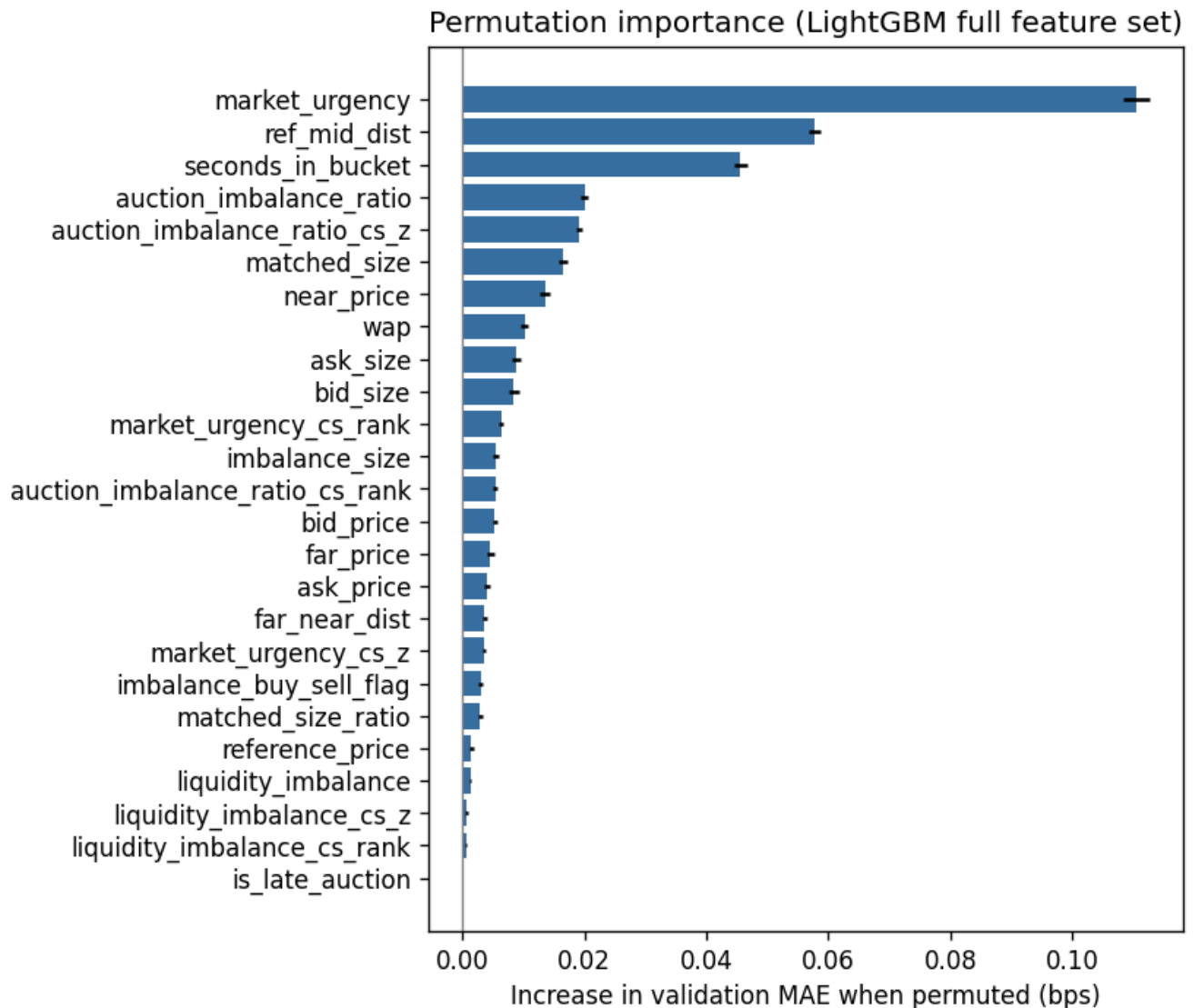


- All five features lower the MAE. The wap move is the biggest contributor. wap_delta and wap_return are near duplicate features, probably because prices sit near one, so a raw and a percentage move are nearly the same features
- There is very little redundancy (except between wap_return and wap_delta), since the individual gains (counting the price feature once) sum up to almost the same value as the gain by all the features. Hence the features mostly carry separate information
- The change in imbalance adds to the level of imbalance. The correlation goes from 0.014 to 0.030 and the MAE falls from 6.4859 to 6.4841
- So the change in imbalance is a better signal than the level of imbalance and it adds to it, but the gain is small and the recent price move matters more. Hence H4 is only partly supported

Feature Importance Diagnostics

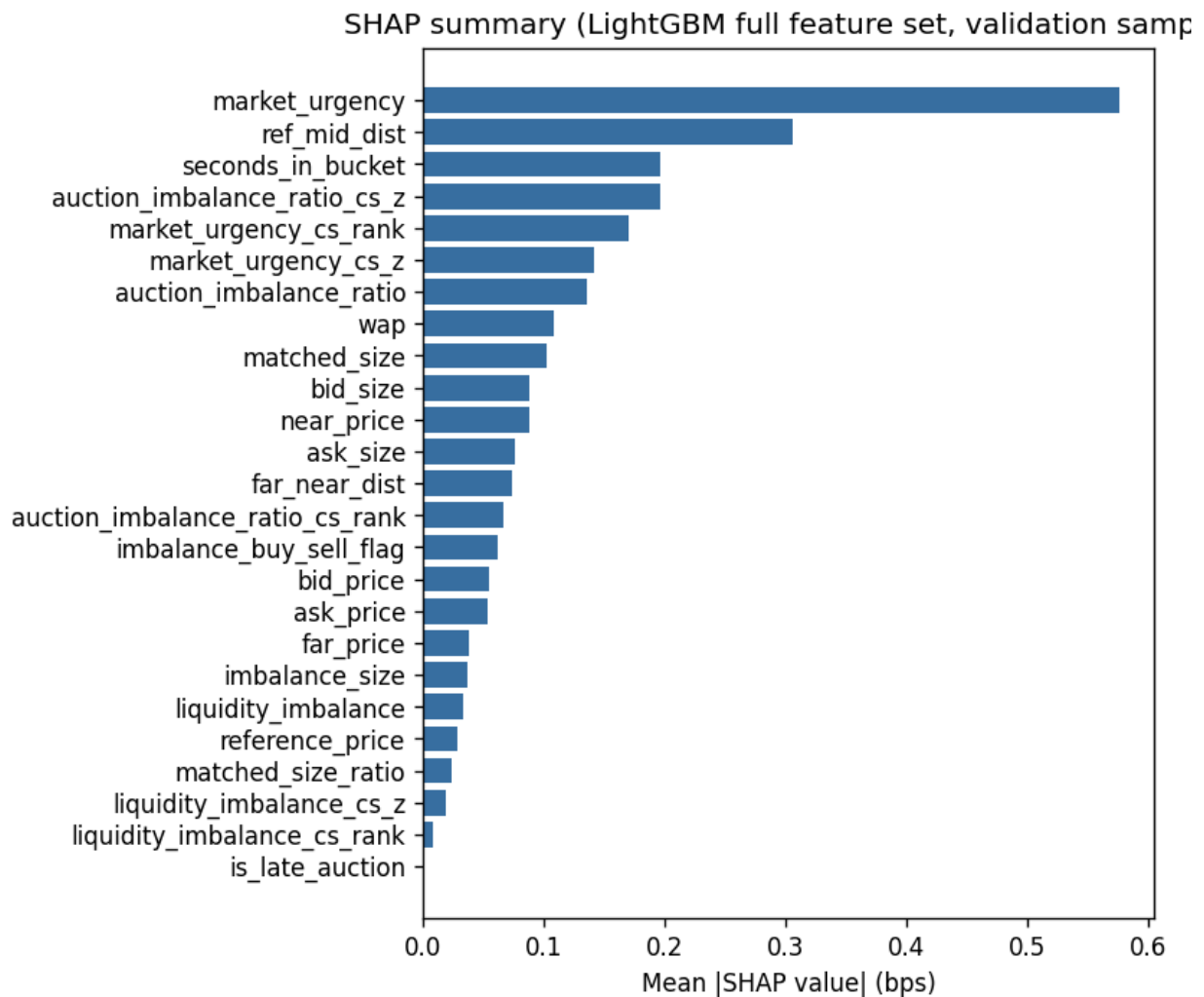
- We first select a random subset of rows. This is done because SHAP can be very expensive for large number of rows and hence we will apply SHAP on randomly chosen small number of rows
- We then create groups of highly related signals (signal + cs_rank + cs_z), this is done because the model might distribute the importance among all the features from related group and the importance might underestimate how much the signal actually matters
- We split the stocks in quantile bins according to their liquidity which is calculated using the mean of matched_size over train data
- [Permutation importance](#) - measures the importance of feature by observing how much the model's performance drops when that feature's values are randomly permuted (in our case we use MAE as the measure of model performance)
- [SHAP \(SHapley Additive exPlanations\)](#) - measures the contribution of each feature to the model using cooperative game theory: $\hat{y} = E[\hat{y}] + \text{sum of contribution of each feature}$
- We report the following:
 - 1: Permutation Importance
 - 2: SHAP importance
 - 3: SHAP through time
 - 4: SHAP across liquidity regimes
- Since SHAP is additive we can add the SHAP for the signal groups created above for a better comparison. On the other hand, permutation importance shares the credit among correlated features
- We wish to answer the following questions
 - Gain importance vs Permutation importance (answered by 1)
 - How much did each feature contribute to each individual prediction (answered by 2)
 - Does the information used by the model change with time (answered by 3)
 - Does the model use different information for liquid & illiquid stocks (answered by 4)

- 1: Permutation Importance vs Gain Importance - Which features actually matter



- The gain method and permutation method broadly agree (spearsman $\rho = 0.84$)
- As expected, the high cardinality rank features perform better in gain than permutation. For example, `market_urgency_cs_rank` gets a 5th position in gain importance whereas 11th position in permutation importance. Similar behaviour is observed for `auction_imbalance_ratio_cs_rank` (7 vs 13)
- The `auction_imbalance_ratio` is undervalued by gain (gain rank 12 vs permutation rank 4). Similar behaviour is observed for `near_price` and `wap`

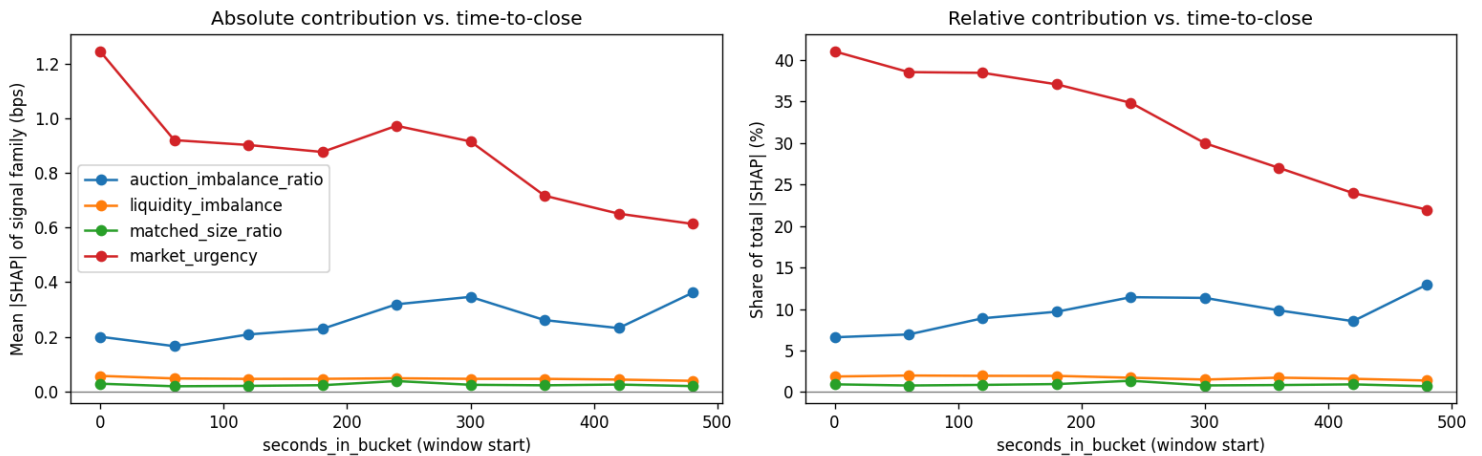
- 2: SHAP importance



- The permutation method and SHAP agree even closely (spearman $\rho = 0.87$)
- Market urgency is the single most important feature. If we look at the complete feature group of market urgency (market urgency + cs_rank + cs_z), it has nearly 33% SHAP share
- Auction imbalance feature group (auction imbalance ratio + cs_rank + cs_z) is the second most dominant family with around 15% SHAP share
- Reference mid distance also emerges as an important feature in gain, permutation and SHAP
- Auction imbalance is informative but not dominant. Hence H1 stands, with the caveat that imbalance is one of the several informative features
- Market urgency works mainly because of the WAP effect explained in RQ1 (WAP moves back towards the mid price) and not because it shows where the real price is going

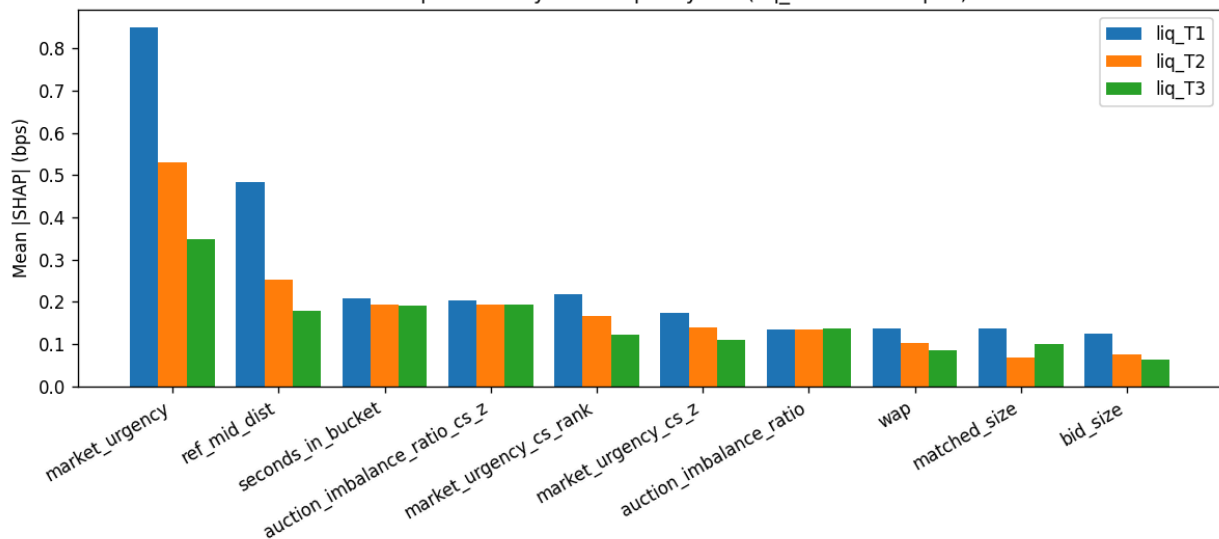
- 3: SHAP through time: Does the information used by the model change with time

SHAP importance of imbalance signals vs. time-to-close



- The contribution of market urgency is maximum at the beginning and fades as time passes. Both its absolute contribution and relative contribution halves. There is a very small bump at around 300 seconds in the absolute contribution.
 - Both the absolute and relative contribution of auction imbalance ratio almost doubles by the end. There is a similar bump at around 300 seconds.
 - As the close approached the information shifts from order book to auction book
 - The contribution of the other two features are negligible and hence can be ignored
- 4: SHAP across liquidity regimes: Does the model use different information for liquid & illiquid stocks

SHAP importance by stock-liquidity tier (liq_T1 = least liquid)

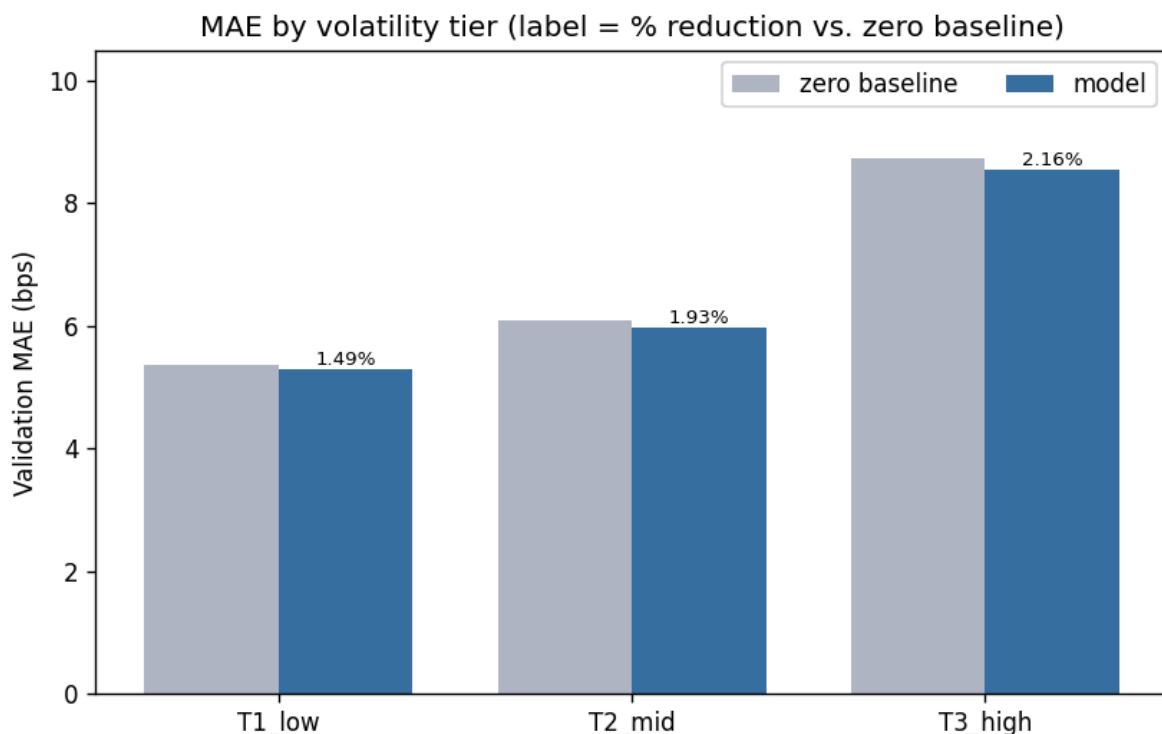


- T1 is the least liquidity tier and T3 is the highest liquidity tier
- The importance is much larger in low liquidity stocks compared to higher liquidity stocks. market_urgency and ref_mid_distance shrink to almost half of their values when we go from T1 to T3

- Auction imbalance is flat across the liquidity tiers. This implies that its relative contribution increases for liquid stocks

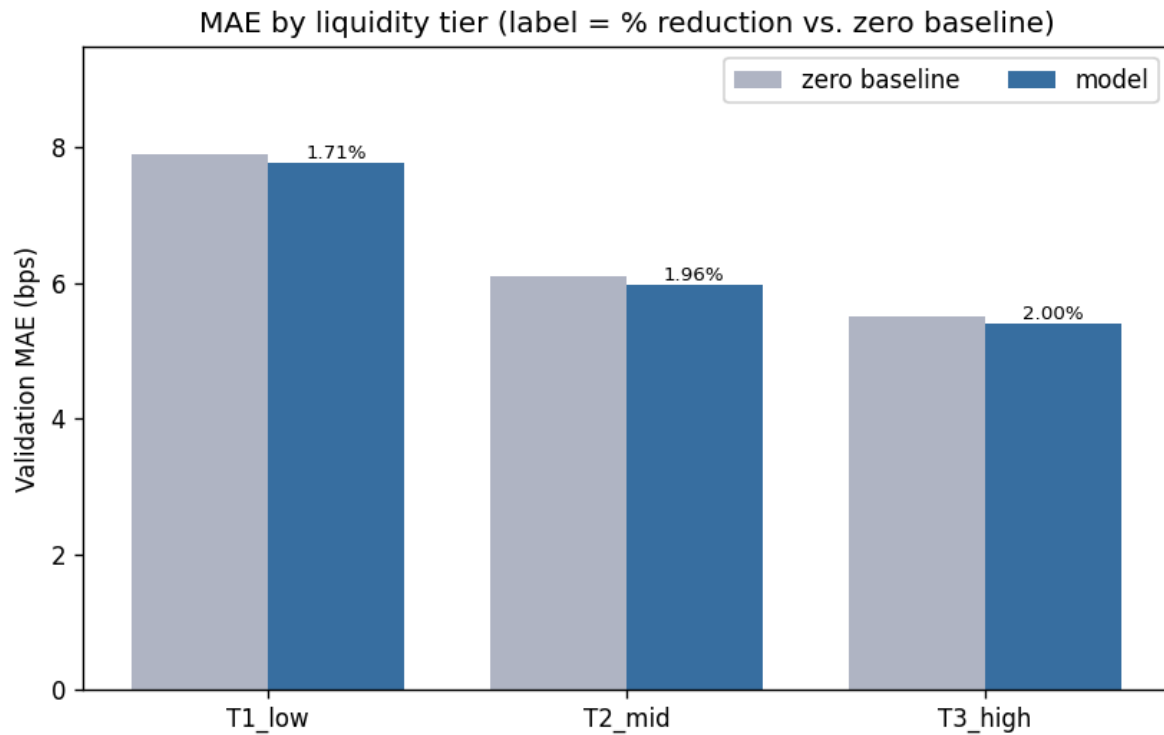
H5 - Regime Analysis - does predictability depend on the market regime

- Regime 1 - Volatility: realized_vol of the last time bucket where it exists. If there is no valid realized_vol, then standard deviation of wap_return over the whole auction period is used instead.
- Regime 2 - Liquidity: mean of matched_size over the whole auction period
- Regime 3 - Imbalance magnitude: mean of |auction_imbalance_ratio| over the whole auction period
- Every auction (one stock on one day) is put into a low, mid or high group for each regime
- The limits of the groups are calculated on the training data and then applied to the validation data
- We use the full LightGBM model and check its performance (correlation, directional accuracy and MAE) separately in each group. MAE cannot be compared directly across groups because the target moves more in some groups. So we also compare the MAE of the model with the MAE of predicting zero in the same group (MAE reduction %)
- To check if the difference between the high and low group is real, we compare the two groups date by date on the validation data
- Volatility



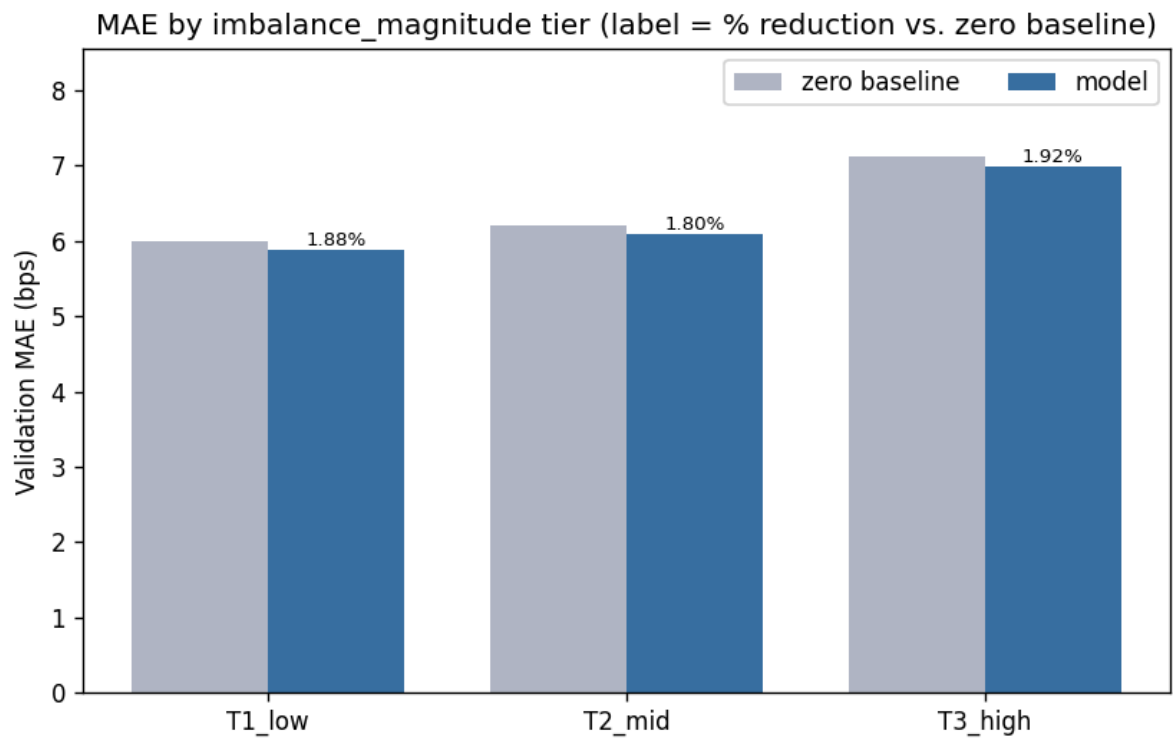
- The MAE goes up from 5.29 in the low volatility group to 8.55 in the high volatility group. This is only because the target itself moves more when volatility is high (the MAE of predicting zero goes up by the same amount)

- The model does slightly better in volatile auctions. The MAE reduction goes from 1.5% to 2.2% and the directional accuracy from 55.0% to 56.4%
- Liquidity



- The MAE is highest for the least liquid group (7.77) and lowest for the most liquid group (5.39), again because the target moves more for illiquid stocks
- The model does slightly worse in the least liquid group. The correlation is 0.166 compared to 0.184 and 0.183 in the mid and high liquidity groups

- Imbalance magnitude



- There is no clear difference between the groups. The correlation is between 0.168 and 0.187 and the MAE reduction between 1.8% and 1.9% in all three groups
- The size of the imbalance does not change how well the model works